

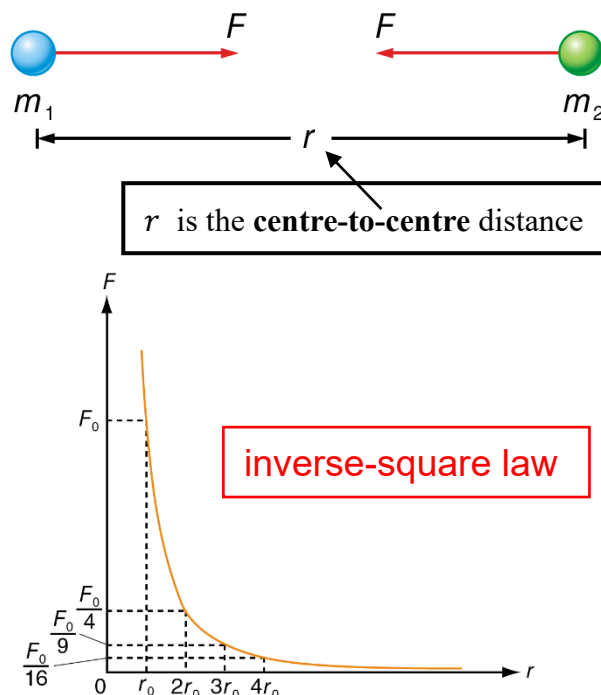
Newton's law of universal gravitation

- Gravitational force between two particles is **directly proportional** to the **product of their masses** ($m_1 \times m_2$) and **inversely proportional** to the **square of the distance** between them (r^2).

$$F = \frac{Gm_1m_2}{r^2}$$

where $G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$
(Universal gravitational constant)

- Gravitational force obeys the **inverse square law**.
- The two forces form an **action and reaction pair**.
- Gravitational force must be **attractive**.



- The mass of the moon is $7.4 \times 10^{22} \text{ kg}$. The mass of the earth is $5.96 \times 10^{24} \text{ kg}$. The average distance from the centre of the earth to the centre of the moon is taken to be $3.8 \times 10^8 \text{ m}$. Estimate the magnitude of the gravitational force between the earth and the moon. Take $G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$.

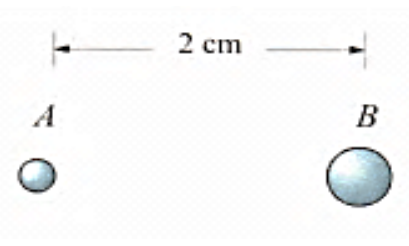
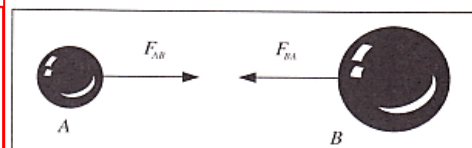
$$F = \frac{GMm}{r^2} \quad F = \frac{6.67 \times 10^{-11} (7.4 \times 10^{22}) (5.96 \times 10^{24})}{(3.8 \times 10^8)^2} = 2.04 \times 10^{20} \text{ N}$$

- Sphere A of mass 100 kg is placed at a distance of 2.0 cm from sphere B of mass 400 kg . Take $G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$. What is the size and direction of the gravitational force of

(a) A on B

(b) B on A

The gravitational pulls on the two masses are equal and opposite, by Newton's 3rd Law. i.e. $F_{AB} = F_{BA}$



By $F = \frac{Gm_Am_B}{r^2}$,

$$F_{AB} = F_{BA} = \frac{6.67 \times 10^{-11} \times 100 \times 400}{0.02^2} = 6.7 \text{ mN}$$

- The gravitational force exerted on the Earth by the Sun is F_0 . The gravitational force exerted on the Sun by the Earth is
 - equal to F_0 and in the same direction.
 - equal to F_0 and in the opposite direction.
 - much smaller than F_0 and in the same direction.
 - much smaller than F_0 and in the opposite direction.

B

These two forces are action and reaction pair, thus, they must be equal in magnitude but opposite in direction.

4. A spaceship at a distance r from the centre of a star experiences a gravitational force F . The spaceship moves a distance $r/2$ towards the star. What is the gravitational force acting on the spaceship when it is at this new location?

- A. $F/2$
B. F
C. $2F$
D. $4F$

Let M and m be the mass of the star and the spaceship respectively.

At distance $= r$, the force is $F = \frac{GMm}{r^2}$.

Thus, at distance $= r/2$, the force is

$$F' = \frac{GMm}{(\frac{r}{2})^2} = 4 \times \frac{GMm}{r^2} = 4F$$

D

5. Two isolated spheres, each of mass m and radius r , are placed in contact and attract each other with a gravitational force F . If one sphere is moved away through a distance of $3r$, the other being kept stationary, what is the new gravitational force between the spheres?

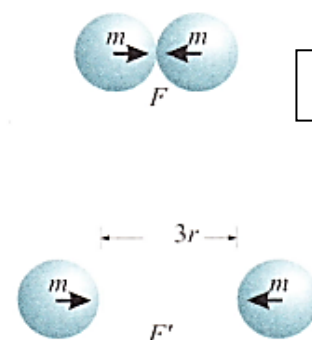
- A. $F/16$
B. $4F/25$
C. $4F/9$
D. $2F/5$

When the spheres are in contact, the separation between the centres of the spheres is not zero but is equal to $2r$. Thus,

$$F = \frac{Gm^2}{(2r)^2} = \frac{Gm^2}{4r^2} \dots\dots\dots(1)$$

When a sphere is moved by $3r$, the separation between their centres is $5r$.

$$F' = \frac{Gm^2}{(5r)^2} = \frac{Gm^2}{25r^2} = \frac{4}{25} \cdot \frac{Gm^2}{4r^2} \\ = \frac{4F}{25}$$



B

6. Two identical spheres, each of mass m and radius r are in contact. One sphere is displaced by a distance $4r$, along the line of centres, away from the first sphere. What is the ratio of the original gravitational force between the sphere to the final gravitational force between them?

- A. 3 : 1
B. 9 : 1
C. 16 : 1
D. 25 : 1

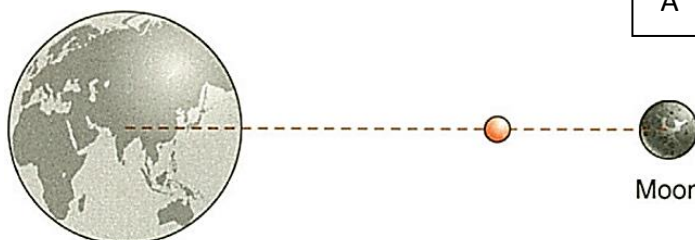
$$F = \frac{G M m}{r^2} \propto \frac{1}{r^2}$$

$$\frac{F_1}{F_2} = \left(\frac{r_2}{r_1}\right)^2 = \left(\frac{r+4r+r}{r+r}\right)^2 = 9$$

B

7. The distance between the Earth and the Moon is 3.78×10^8 m. An object is placed along the line joining the centres of the two bodies and is located at a distance of 3.78×10^7 m from the Moon. If the object experiences zero gravitational force, what is the ratio of the mass of the Moon to that of the Earth.

- A. 1 : 81
B. 1 : 90
C. 1 : 110
D. 1 : 333



A

The gravitational force acted by the Earth is equal to that acted by the Moon, i.e.

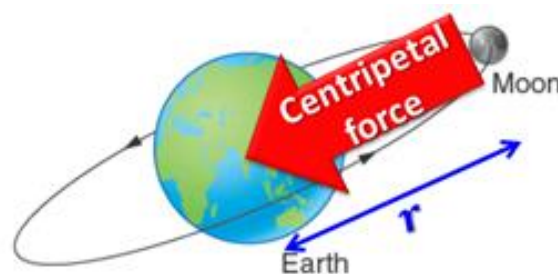
$$\frac{GM_E m}{(R_{ME} - r)^2} = \frac{GM_M m}{r^2} \\ \frac{M_M}{M_E} = \frac{r^2}{(R_{ME} - r)^2} = \frac{(3.78 \times 10^7)^2}{(3.78 \times 10^8 - 3.78 \times 10^7)^2} = \frac{1}{81}$$

Circular motion under gravity

- In space, **gravitational force** between two objects provide the **required centripetal force** to keep object in **uniform circular motion**.

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$\therefore v = \sqrt{\frac{GM}{r}}$$



Remarks: Speed of orbiting object depend on the mass of central massive body M but not mass of the object m .

By $\frac{GMm}{r^2} = mr\omega^2$ and $\omega = \frac{2\pi}{T}$

$$\therefore r^3 = \frac{GM}{4\pi^2} T^2$$

Remarks: An object farther away from the Earth moves with a lower speed and takes longer time to orbit the Earth.

- Since gravitational force is always perpendicular to the movement of the object, no work is done by the gravitational force. Hence, the object moves around the earth with constant speed and kinetic energy.

8. 2018 DSE MC Q.13

A satellite of mass m moves around a planet of mass M in a circular orbit of radius r . What does the angular velocity of the satellite depend on ?

- (1) r
(2) m
(3) M
A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

The centripetal force of the satellite is given by the gravitational force acted by the planet.

$$\therefore m\omega^2 r = G \frac{Mm}{r^2} \Rightarrow \omega = \sqrt{\frac{GM}{r^3}}$$

C

9.

Long ago, astronomers wanted to find out whether the ring of Saturn is a rigid body or a group of satellites revolving around Saturn. The linear speeds v of different layers of the ring were measured and how v varies with the distance r from the centre of Saturn was determined. Which of the following relations between v and r supports the suggestion that the ring is a group of satellites revolving around Saturn ?

- A. $v \propto r$
B. $v \propto \sqrt{r}$
C. $v \propto \frac{1}{r}$
D. $v \propto \frac{1}{\sqrt{r}}$

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \therefore v^2 = \frac{GM}{r}$$

$$\therefore v \propto \frac{1}{\sqrt{r}}$$

D

10. Assume that the Moon revolves around the Earth in a circular orbit of radius $3.78 \times 10^8 \text{ m}$. The mass of the earth is $5.96 \times 10^{24} \text{ kg}$. What is the orbital speed of the moon? (Take $G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$)

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{6.67 \times 10^{-11} \times 5.96 \times 10^{24}}{3.78 \times 10^8}} = 1025.5 \text{ ms}^{-1} \sim 1030 \text{ ms}^{-1}$$

11. Assume the earth circulates the sun in a circular orbit. If the period of one complete circulation is 365 days, then what is the radius of the orbit? (Take $G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$ and Mass of sun = $2 \times 10^{30} \text{ kg}$)

- A. $1.5 \times 10^{11} \text{ m}$
B. $2.19 \times 10^{11} \text{ m}$
C. $2.76 \times 10^{11} \text{ m}$
D. $5.8 \times 10^{11} \text{ m}$

Let M be the mass of the Sun and m be the mass of the Earth.

$$\frac{GMm}{r^2} = mr\omega^2$$

$$r^3 = \frac{GM}{\omega^2} = \frac{GM}{\left(\frac{2\pi}{T}\right)^2} = \frac{GMT^2}{4\pi^2}$$

$$r = \sqrt[3]{\frac{(6.67 \times 10^{-11})(2 \times 10^{30})(365 \times 24 \times 3600)^2}{4\pi^2}} = 1.5 \times 10^{11} \text{ m}$$

A

12. A satellite moves around the Earth in a circular orbit at a height 1200 km above the Earth's surface. Estimate the linear speed of the satellite. Take the mass and the radius of the Earth to be $5.98 \times 10^{24} \text{ kg}$ and 6370 km respectively.

- A. 7260 m s^{-1}
B. 8910 m s^{-1}
C. 18200 m s^{-1}
D. 236000 m s^{-1}

A Gravitational force = centripetal force

$$\frac{GM_E m}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{GM_E}{r}}$$

$$= \sqrt{\frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24}}{6.37 \times 10^6 + 1.20 \times 10^6}} = 7260 \text{ m s}^{-1}$$

A

13. The earth is at a distance r metres from the centre of the sun. It takes 365 days for the earth to revolve once around the sun in a circular path. Find the mass of the sun in terms of r . (Neglect the effects of other planetary objects.)

- A. $1.95 \times 10^{-4} r^3 \text{ kg}$
B. $5.95 \times 10^{-4} r^3 \text{ kg}$
C. $1.85 \times 10^4 r^3 \text{ kg}$
D. $1.25 \times 10^6 r^3 \text{ kg}$

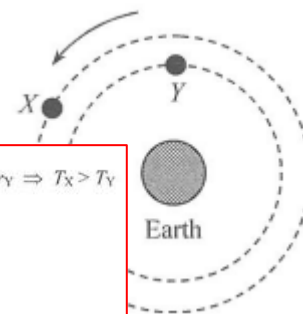
$$\text{By } \frac{GMm}{r^2} = mr\omega^2 \quad \therefore \frac{GM}{r^3} = \omega^2 = \left(\frac{2\pi}{T}\right)^2$$

$$\therefore M = \frac{4\pi^2 r^3}{GT^2} = \frac{4\pi^2 r^3}{(6.67 \times 10^{-11}) \cdot (365 \times 24 \times 3600)^2} = 5.95 \times 10^{-4} r^3$$

B

14. Two identical satellites X and Y are moving in two circular orbits around the Earth as shown. Which statement is/are correct?

- (1) The period of X is greater than that of Y .
(2) The speed of X is smaller than that of Y .
(3) The gravitational force on X is smaller than that on Y .



- A. (1) & (2) ✓
B. (1) & (3) ✓
C. (2) & (3) ✓
D. (1), (2) & (3) ✓
- (1) $\frac{GMm}{r^2} = mr\omega^2 \quad \therefore \frac{GM}{r^3} = \omega^2 = \left(\frac{2\pi}{T}\right)^2 \quad \therefore T^2 \propto r^3 \quad \therefore r_X > r_Y \Rightarrow T_X > T_Y$
(2) $\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \therefore v^2 = \frac{GM}{r} \quad \therefore r_X > r_Y \Rightarrow v_X < v_Y$
(3) $F = \frac{GMm}{r^2} \propto \frac{1}{r^2} \quad \therefore r_X > r_Y \Rightarrow F_X < F_Y$

D

15. Two satellites A and B circle round the earth with the orbitals of radius r and $2r$ respectively. If the orbital speed of satellite A and that of satellite B are v_1 and v_2 respectively. Find the ratio v_1 to v_2 .

- A. 2 : 1
B. $\sqrt{2} : 1$
C. 1 : 1
D. $1 : \sqrt{2}$

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$v \propto \frac{1}{\sqrt{r}}$$

Hence,

$$\therefore \frac{v_1}{v_2} = \sqrt{\frac{r_2}{r_1}} = \sqrt{\frac{2r}{r}} = \sqrt{2}$$

B

16. The radius of the Earth is R . Satellite X orbits around the Earth at a height of R , while satellite Y orbits around the Earth at a height of $2R$. Find the ratio of the speed of X to that of Y .

- A. $\sqrt{\frac{1}{2}}$
B. $\sqrt{\frac{2}{1}}$
C. $\sqrt{\frac{2}{3}}$
D. $\sqrt{\frac{3}{2}}$

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \therefore v \propto \sqrt{\frac{1}{r}}$$

$$\therefore \frac{v_X}{v_Y} = \sqrt{\frac{r_Y}{r_X}} = \sqrt{\frac{(R+2R)}{(R+R)}} = \sqrt{\frac{3}{2}}$$

D

17. Two satellites A and B are moving around the earth in circular orbits. The speed of A is twice of that of B. What is the ratio of the radius of the orbit of A to that of B?

- A. 4 : 1
B. 1 : 4
C. 2 : 1
D. 1 : 2

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow GM = v^2 r = \text{constant}$$

$$v_A^2 r_A = v_B^2 r_B$$

$$\left(\frac{v_A}{v_B}\right)^2 = \frac{r_B}{r_A}$$

$$\left(\frac{2}{1}\right)^2 = \frac{r_B}{r_A}$$

$$\frac{r_A}{r_B} = \frac{1}{4}$$

B

18. One of Jupiter's moons is called Ganymede. The radius of Ganymede's orbit around Jupiter is about 3 times that of the Moon around the Earth. The mass of Jupiter is 318 times that of the Earth. If the period of the Moon around the Earth is 27.3 days, estimate the period of Ganymede revolving around Jupiter.

- A. 2.7 days
B. 8.0 days
C. 91 days
D. 273 days

The satellite moves round the planet in circular motion, thus the gravitational force provides the centripetal force.

As $\frac{GMm}{r^2} = mr\omega^2$

$$\therefore \frac{GM}{r^3} = \omega^2 = \left(\frac{2\pi}{T}\right)^2 \quad \therefore T^2 = \frac{4\pi^2 r^3}{GM} \propto \frac{r^3}{M}$$

$$\therefore \left(\frac{T_1}{T_2}\right)^2 = \left(\frac{r_1}{r_2}\right)^3 \left(\frac{M_2}{M_1}\right)$$

$$\therefore \left(\frac{T_1}{27.3}\right)^2 = \left(\frac{3}{1}\right)^3 \left(\frac{1}{318}\right)$$

$$\therefore T_1 = 8.0 \text{ days}$$

B

Gravitational field strength

- Gravitational field strength g is the **gravitational force acting per unit mass** at a point.

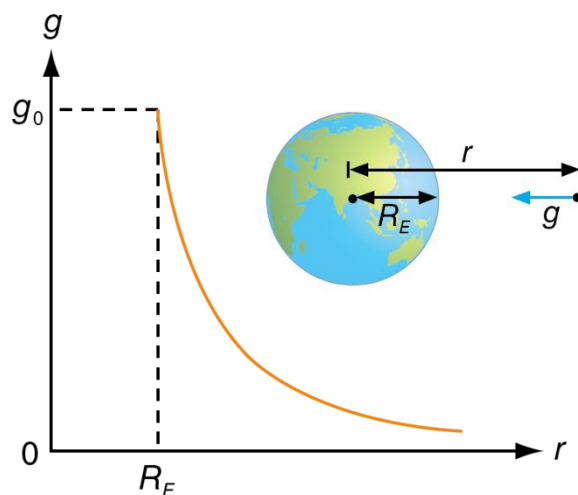
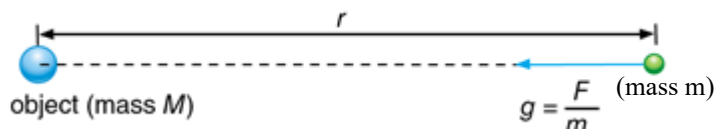
$$g = \frac{F}{m}$$

(unit: Nkg^{-1} or ms^{-2})

By $F = \frac{GMm}{r^2}$ and $g = \frac{F}{m}$

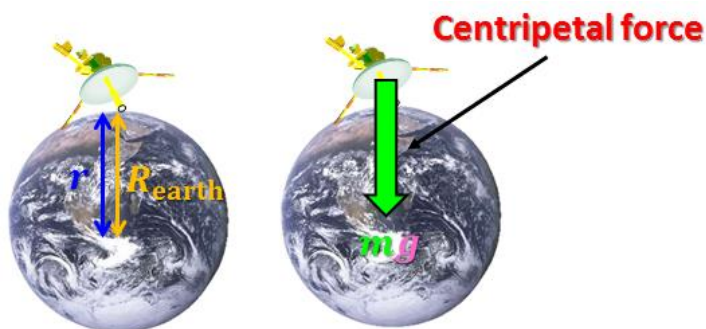
$$\therefore g = \frac{GM}{r^2}$$

- g is also called the acceleration due to gravity.
- Gravitational field strength due to central mass M also obey inverse square law.



Low altitude orbit (near earth's surface)

- A low altitude satellite moves around the earth in an orbit **very close** to the Earth's surface. The height h of the satellite above the Earth's surface is negligible compared to Earth's radius.
- Since the satellite is close to the Earth's surface, the acceleration of the satellite is equal to the gravitational field strength g at the Earth's surface.



$$mg = \frac{mv^2}{r} = mr\omega^2$$

19. What is the speed of a low altitude satellite near the earth's surface?

(Given: $R_{\text{earth}} = 6.37 \times 10^6 \text{ m}$, $g = 9.81 \text{ ms}^{-2}$)

$$\frac{GM}{r^2} = \frac{v^2}{r} \quad 9.81 = \frac{v^2}{6.37 \times 10^6} \quad r = 7905 \text{ ms}^{-1}$$

20. Given that the Earth's radius is 6400 km and the gravity on the Earth's surface is 9.8 ms^{-2} . Estimate the period of an artificial Earth-orbiting satellite that passes just above the Earth's surface.

Since the satellite is orbiting close to the Earth's surface, its orbital radius is $R_e = 6.4 \times 10^6 \text{ m}$ and the centripetal force is equal to (provided by) its weight, mg . So, we have

$$mg = mR_e \omega^2$$

$$\therefore \omega = \sqrt{\frac{g}{R_e}}$$

The period of the satellite is therefore

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{R_e}{g}} = 2\pi \sqrt{\frac{6.4 \times 10^6}{9.8}}$$

$$= 5078 \approx \underline{5.1 \times 10^3 \text{ s}}$$

21. A weather satellite orbits the earth 14 times per day. Determine its height above the earth surface. Take $GM_{\text{earth}} = 4 \times 10^{14} \text{ J m kg}^{-1}$ and $R_{\text{earth}} = 6.4 \times 10^6 \text{ m}$

The radius r of the orbit is given by

$$\frac{GM_e m}{r^2} = mr\omega^2$$

$$r^3 = GM_e \left(\frac{T}{2\pi} \right)^2$$

$$\therefore r = \sqrt[3]{4.0 \times 10^{14} (6.2 \times 10^3 / 2\pi)^2} = 7.28 \times 10^6 \text{ m}$$

So its height above the earth is $8.8 \times 10^5 \text{ m}$

The period of the weather satellite is $T = \frac{24 \times 3600}{14} = 6171 \text{ s}$

22. The moon moves around the earth with a mean radius of $3.8 \times 10^8 \text{ m}$ and a period of 27.3 days. Estimate the radius of the orbit of a satellite if it circles round the with a period of 1 week.

C

$$T^2 \propto r^3$$

$$\frac{(1 \times 7)^2}{27.3^2} = \frac{r^3}{(3.8 \times 10^8)^3}$$

$$\therefore r = \sqrt[3]{\frac{7^2}{27.3^2} \times (3.8 \times 10^8)^3} = 1.5 \times 10^8 \text{ m}$$

23. A satellite orbits around the Earth in a circular orbit at a height $3.8 \times 10^5 \text{ m}$ above the Earth's surface. What is the period of the satellite? (Given: radius of the Earth = 6370 km, mass of the Earth = $5.98 \times 10^{24} \text{ kg}$)

Gravitational force = centripetal force

$$\frac{GM_E m}{r^2} = mr\omega^2$$

$$\frac{GM_E}{(R_E + h)^2} = (R_E + h) \left(\frac{2\pi}{T} \right)^2$$

$$T = 2\pi \sqrt{\frac{(R_E + h)^3}{GM_E}}$$

$$= 2\pi \sqrt{\frac{(6.37 \times 10^6 + 3.8 \times 10^5)^3}{6.67 \times 10^{-11} \times 5.98 \times 10^{24}}}$$

$$= 5520 \text{ s}$$

$$= 1 \text{ hr } 32 \text{ min}$$

24. A satellite revolves around and very close to the Earth in a circular orbit with a constant speed. The radius of the orbit is 6.8×10^6 m. The period of revolution is 1.55 hours. Estimate the radius of the Earth (The universal gravitational constant G to be $6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$)

a) Gravitational force = centripetal force

$$\frac{GM_E m}{r^2} = m r \omega^2 \quad 1M$$

$$\frac{GM_E}{R_E^2} \times \frac{R_E^2}{r^2} = r \left(\frac{2\pi}{T} \right)^2 \quad 1M$$

$$R_E = \sqrt{\frac{4\pi^2 r^3}{g T^2}} \quad 1M$$

$$= \sqrt{\frac{4\pi^2 (6.8 \times 10^6)^3}{9.81 \times (1.55 \times 60 \times 60)^2}}$$

$$= 6.3749 \times 10^6 \text{ m} \approx 6.37 \times 10^6 \text{ m} \quad 1A$$

$\therefore$ The radius of the Earth is 6.37×10^6 m.

25. A satellite revolves around the Earth in a circular orbit with a period of 120 minutes. The gravitational field strength on the Earth's surface is 9.81 N kg^{-1} and the radius of the Earth is 6370 km. What is the height of the satellite above the Earth's surface?

Gravitational force = centripetal force

$$\frac{GM_E m}{r^2} = m r \omega^2$$

$$\frac{GM_E}{R_E^2} \times \frac{R_E^2}{r^2} = r \left(\frac{2\pi}{T} \right)^2$$

$$r = \sqrt{\frac{g T^2 R_E^2}{4\pi^2}}$$

MISCONCEPTION
Some students do not realise that the gravitational field strength at the orbit is smaller than the field strength at the Earth's surface. Some can think that the field strength there is the same as at the Earth's surface.

$$= \sqrt{\frac{9.81 \times (120 \times 60)^2 \times (6.37 \times 10^6)^2}{4\pi^2}} = 8.06 \times 10^6 \text{ m}$$

Height above the Earth's surface = $8.06 \times 10^6 - 6.37 \times 10^6 = 1.69 \times 10^6 \text{ m}$

26. Find the gravitational field strength at a point which is 1610 km above the Mars's surface. (Given: Mass of Mars $M_M = 6.42 \times 10^{23} \text{ kg}$, Radius of Mars $R_M = 3390 \text{ km}$, $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$)

$$g = \frac{GM}{r^2}$$

$$g = \frac{(6.67 \times 10^{-11})(6.42 \times 10^{23})}{(1610 \times 10^3 + 3390 \times 10^3)^2} = 1.71 \text{ N kg}^{-1}$$

27. Find the gravitational field strength at an altitude of $2R$ above the earth's surface where R is the radius of the earth.

$$g_r = \frac{GM}{(3R)^2} = \frac{gR^2}{9R^2} = \frac{1}{9} \times 10 = 1.11 \text{ m s}^{-2}$$

28. The ratio of the masses of planets X and Y is $5 : 1$ and the ratio of their radii is $3 : 1$. Find the ratio of the gravitational field strength on the surface of X to that of Y .

B Given: $\frac{M_X}{M_Y} = 5$ and $\frac{R_X}{R_Y} = 3$

By $g = \frac{GM}{r^2}$, $g \propto \frac{M}{r^2}$

$$\frac{g_X}{g_Y} = \frac{M_X R_Y^2}{M_Y R_X^2} = 5 \times \left(\frac{1}{3} \right)^2 = \frac{5}{9}$$

Parking / Geostationary orbit

- A parking satellite always stays at rest above a certain position on the Earth's surface.
- The satellite must be **equatorial**.
- The satellite must orbit with the **same period** as the Earth ($T = 24$ hours).
- The satellite must orbit in the same direction as the Earth's rotation.



29. Calculate the radius of a parking orbit. ($G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$, $M_{\text{earth}} = 5.98 \times 10^{24}$)

$$v = \frac{2\pi r}{T}, \quad v = \sqrt{\frac{GM_{\text{earth}}}{r}}, \quad \frac{2\pi r}{T} = \sqrt{\frac{GM_{\text{earth}}}{r}}$$

$$\frac{4\pi^2 r^2}{T^2} = \frac{GM_{\text{earth}}}{r}, \quad r^3 = \frac{GM_{\text{earth}} T^2}{4\pi^2}$$

$$r^3 = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(24 \times 60 \times 60)^2}{4\pi^2} = 4.23 \times 10^7 \text{ m}$$

30. Which of the following statements about the geostationary orbit of a satellite over the earth is/are true?

- The geostationary orbit is very close to the earth
- The radius of the geostationary orbit is independent of the mass of the satellite.
- The only geostationary orbit is in the plane of the equator.

- (1) only
- (3) only
- (1) and (2) only
- (2) and (3) only

D

31. A geostationary satellite travels in a circular orbit around the Earth. The radius of the Earth is 6370 km. Which of the following statements about the satellite is/are correct?

- It can be placed over any place on the Earth.
- Its period is 8.64×10^4 s.
- It's 3.59×10^7 m above the ground.

- (1) only
- (2) only
- (2) and (3) only
- All of the above

C

C A geostationary satellite must be placed over the Earth's Equator. $\therefore$ (1) is incorrect.
A geostationary satellite has the same period as the Earth, i.e. 24 hours.
 $T = 24 \times 60 \times 60 = 8.64 \times 10^4$ s
 $\therefore$ (2) is correct.

MISCONCEPTION
Some students fail

Gravitational force = centripetal force

$$\frac{GM_E m}{r^2} = m r \omega^2$$

$$\frac{GM_E}{R_E^2} \times \frac{R_E^2}{r^2} = r \left(\frac{2\pi}{T} \right)^2$$

$$r = \sqrt[3]{\frac{g T^2 R_E^2}{4\pi^2}}$$

$$= \sqrt[3]{\frac{9.81 \times (8.64 \times 10^4)^2 \times (6.37 \times 10^6)^2}{4\pi^2}}$$

$$= 4.222 \times 10^7 \text{ m}$$

Height above the Earth's surface
 $= 4.222 \times 10^7 - 6.37 \times 10^6$
 $= 3.59 \times 10^7 \text{ m}$

$\therefore$ (3) is correct.

32. A satellite revolves around the Earth in a circular orbit 8 times per day. The radius of the Earth is 6370 km. The mass of the satellite is 250 kg.

- (a) Find the period of the satellite. (1 mark)
(b) Find the orbital radius of the satellite. (4 marks)
(c) Find the kinetic energy of the satellite. (3 marks)

(a) Period of the satellite = $\frac{24 \times 60 \times 60}{8} = 10\,800\text{ s}$

1A

(b) Gravitational force = centripetal force

$$\begin{aligned}\frac{GM_E m}{r^2} &= m r \omega^2 \\ \frac{GM_E}{R_E^2} \times \frac{R_E^2}{r^2} &= r \left(\frac{2\pi}{T} \right)^2 \\ r &= \sqrt[3]{\frac{g T^2 R_E^2}{4\pi^2}} \\ &= \sqrt[3]{\frac{9.81 \times 10\,800^2 \times (6.37 \times 10^6)^2}{4\pi^2}} \\ &= 1.0555 \times 10^7\text{ m} \\ &\approx 1.06 \times 10^7\text{ m}\end{aligned}$$

1M

1M

1M

1A

$\therefore$ The orbital radius of the satellite is $1.06 \times 10^7\text{ m}$.

(c) Kinetic energy of the satellite = $\frac{1}{2} m v^2$

$$\begin{aligned}&= \frac{1}{2} m \left(\frac{2\pi r}{T} \right)^2 \\ &= \frac{1}{2} \times 250 \times \left(\frac{2\pi \times 1.0555 \times 10^7}{10\,800} \right)^2 \\ &= 4.71 \times 10^9\text{ J}\end{aligned}$$

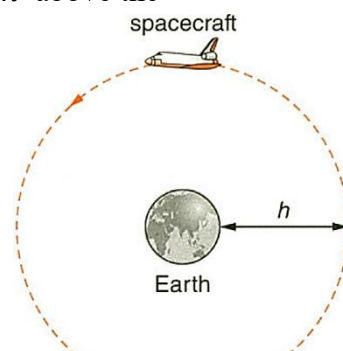
1M

1M

1A

33. A spacecraft travels around the Earth in a circular orbit with a constant speed. The gravitational field strength g' at the position of the spacecraft is 9.4 N kg^{-1} . The spacecraft is h above the surface of the Earth. The radius of the Earth is 6370 km.

- (a) Find h . (3 marks)
(b) At what speed does the spacecraft travel? (2 marks)
(c) Find the period of the spacecraft. (2 marks)
(d) The weight of the spacecraft is 75 200 N when it moves along this orbit. What is its mass? (2 marks)
(e) An astronaut in the spacecraft feels **weightless**. Why? (2 marks)



(a) By $g = \frac{GM}{r^2}$,

$$g' = \frac{GM_E}{R_E^2} \times \frac{R_E^2}{r^2}$$

$$r = R_E \sqrt{\frac{g}{g'}}$$

$$= 6.37 \times 10^6 \times \sqrt{\frac{9.81}{9.4}}$$

$$= 6.507 \times 10^6\text{ m}$$

$$h = 6.507 \times 10^6 - 6.37 \times 10^6$$

$$= 1.37 \times 10^5\text{ m}$$

(b) By $a = \frac{v^2}{r}$,

$$\text{speed of spacecraft} = \sqrt{ar} = \sqrt{g'r} = \sqrt{9.4 \times 6.507 \times 10^6} = 7820\text{ m s}^{-1}$$

(c) Period = $\frac{2\pi r}{v}$

$$= \frac{2\pi \times 6.507 \times 10^6}{7820}$$

$$= 5230\text{ s}$$

(d) By $W = mg$,

$$\text{mass} = \frac{W}{g}$$

$$= \frac{75\,200}{9.4}$$

$$= 8000\text{ kg}$$

- (e). Since the astronaut and the spacecraft move with the same centripetal acceleration, all of astronaut's weight is completely used for centripetal force and no normal reaction is acting on astronaut. Weightlessness is then experienced by the astronaut.